

Area	Detail
CPU	Intel Core 8th Generation - Family 6, Model 142, Stepping 12 (Whiskey Lake, x86-64)
RAM	16 GB (15 GB available to OS)
Operating System	Windows 11, Build 26200
Python Version	3.13.2
Cryptography	46.0.5 ECDH P-256 and RSA-2048 via hazmat primitives
kybercffi	1.0.1 Kyber512 (ML-KEM-512) key encapsulation
pycryptodome	3.23.0 AES-CBC encryption and PKCS7 padding
hashlib	Python standard library SHA-256, SHA3-256, SHAKE256
numpy / scipy / matplotlib	2.4.2 / 1.17.1 / 3.10.8 statistical analysis and visualisation
ML-KEM Variant	Kyber512 (ML-KEM-512, NIST Security Level 1, ~128-bit post-quantum security) [6][17]
AES Mode	CBC (Cipher Block Chaining) consistent across Scenarios 3 and 4
Padding	PKCS7 via Crypto.Util.Padding (pad / unpad)
Randomness	Keys: library CSPRNG (OS-seeded); IV: os.urandom(16); Plaintext (Hybrid/AES): os.urandom(size); Hash input: os.urandom(size), pre-generated outside time block
Timing Function	time.perf_counter() : high-resolution performance counter, sub-microsecond resolution, returns float seconds
Warm-up Iterations	50 iterations executed and discarded before recording begins (Scenarios 1 and 4)
System Controls	Native hardware (no virtualisation); background applications minimised; power plan set to High Performance to prevent CPU frequency scaling

Table IV.1: Complete Experimental Environment Specification

Scenario	Algorithms	Iterations	Input / Workload	Timing Scope
Key Exchange	ECDH P-256 · RSA-2048 · (ML-KEM-512)	2,000 (+50 warm-up)	32-byte key (RSA); SECP256R1; Kyber512 params	See Section IV-E
Hash Functions	SHA-256 · SHA3-256 · SHAKE256	2,000 per algo per size	1 KB and 1 MB; os.urandom(size), pre-generated outside timed block	Single hash_func(data).digest() : SHAKE256: .digest(32)
AES	AES-128 / 192 / 256-CBC	200 per configuration(+50 warm up)	1 KB, 10 MB, 100 MB; os.urandom(size); IV: os.urandom(16)	Full AES-CBC pad + encrypt
Hybrid	ECDH + Kyber512 + SHA-256 KDF + AES-256-CBC	2,000 (+50 warm-up) key est; 200 AES	1 KB / 1 MB AES payload	Each component timed independently

Table IV.2: Benchmark Configuration per Scenario